

Module 2: Network Security Essentials – One-Page Note

1. Introduction

Network Security focuses on protecting networked systems from unauthorized access, misuse, or attacks. As organizations rely heavily on interconnected systems, safeguarding network infrastructure has become a vital part of cyber security.

2. Firewalls

A firewall is a security device (hardware or software) that monitors and controls incoming and outgoing network traffic based on predefined rules.

Types of Firewalls

- **Packet-Filtering Firewall – Checks source/destination IP, port, and protocol**
- **Stateful Firewall – Tracks the state of active connections**
- **Application-Level Gateway – Filters traffic at the application layer**
- **Next-Generation Firewall (NGFW) – Includes intrusion prevention, deep inspection, and advanced threat protection**

Purpose: To block malicious traffic while allowing legitimate communication.

3. Virtual Private Networks (VPNs)

A VPN creates a secure, encrypted tunnel between a user and a network over the internet.

Why VPNs are important:

- **Safeguard data transmitted over public networks**
 - **Ensure confidentiality by encrypting traffic**
 - **Provide remote workers secure access to internal systems**
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4. Intrusion Detection & Prevention Systems (IDS/IPS)

IDS (Intrusion Detection System)

- **Monitors network traffic to identify suspicious activity**
- **Alerts administrators when threats are detected**
- **Works as a passive monitoring tool**

IPS (Intrusion Prevention System)

- **Actively blocks or stops malicious traffic**
 - **Prevents attacks in real-time**
 - **Often integrated with firewalls in modern solutions**
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5. Secure Network Architecture

Designing a network with security in mind reduces vulnerabilities.

Core elements:

- **Segmentation – Dividing the network into isolated sections to limit attack spread**
- **DMZ (Demilitarized Zone) – A separate zone for public-facing servers to protect internal networks**
- **Least Privilege Access – Users only get the access they need**
- **Redundancy – Backup systems to ensure availability**

A secure architecture minimizes entry points and strengthens the organization's overall security posture.

6. Network Protection Techniques

- **Encryption – Protects data during transmission**
 - **Access Control Lists (ACLs) – Define who can access what resources**
 - **MAC Filtering – Restricts devices based on hardware addresses**
 - **Antivirus & Anti-malware – Prevent infections from spreading**
 - **Patch Management – Fixes vulnerabilities in network devices**
 - **Security Monitoring – Continuous surveillance using SIEM tools**
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7. Conclusion

Network Security Essentials provide the foundational mechanisms that protect an organization's communications and data flow. Understanding firewalls, VPNs, IDS/IPS, and secure network design is critical for building strong defense systems against modern cyber threats.

